

Gas Field Policy 18: Sour Gas Service

GENERAL:

Hydrogen Sulfide H_2S is a colorless gas which is harmful at concentrations of 20 ppm in air. At concentrations of 800 ppm death results within 30 minutes due to destruction of lung tissues. At low concentrations (1 to 20 ppm), it has a strong rotten egg odor and is harmless. However, at higher concentrations, the sense of smell is anesthetized and the gas is odorless. It is heavier than air (mole. wt. 34) and so tends to accumulate in low places. Due to the hazardous nature of this gas, compressors handling H_2S should not be approached unless H_2S leak detection equipment is functioning.

Effects on metals: H_2S tends to cause sulfide stress cracking in steels when all of the following four conditions exists:

- A. The H_2S concentration and pressure condition fall into the "sulfide stress cracking region: of NACE MR-0175 (figure 1).
- B. The steel component has a hardness over 22 Rockwell "C".
- C. The steel component is stressed.
- D. Moisture is present in the gas.

Sulfide stress cracking is more aggressive if chlorides are also present in the gas.

To prevent sulfide stress cracking of components that lie within the NACE definition of sour gas service, materials of less than 22 HRC must be used. Alternatively, materials with a greater resistance to attack may be used (e.g. Age Hardened 17-4 PH stainless steel piston rods).

MATERIALS OF CONSTRUCTION:

- A. If the concentration of H_2S , at the discharge pressure of any stage, falls into the "B" range (figure 1), Siemens Energy recommends its standard materials of construction.
- B. If the concentration of H_2S , at the discharge pressure of any stage, falls into the "A" range (figure 1), Siemens Energy requires the following construction for maximum resistance to sulfide stress cracking and containment of the gas.
 - 1) Distance pieces: Distance pieces must be arranged so that potentially harmful leakage of H_2S to the atmosphere cannot occur. Possible alternatives are:
 - A) For all applications Siemens Energy recommends the use of a single compartment distance piece equipped with purged packing case, purged wiper case and slinger if desired (figure 2). The distance piece and packing vents should be piped to a flare or recovery system. In some cases an extra long distance piece may be required.
 - B) Standard single-compartment distance pieces may be used with caution if a sweet gas packing purge and vacuum scavenge pump is used to maintain the distance piece at negative pressure and gas leakage is eliminated. Piston rod oil slingers are recommended to minimize frame oil contamination.
 - C) Two compartment distance pieces are available if preferred. Please consult with the Siemens Energy HSRC Group for further details.

- 2) Bolting:
 - A) Bolting that is in the cylinder end distance piece is outside the NACE definition of sour gas service (less than 65 PSIG). Therefore, Siemens Energy will use standard materials for the distance piece studs, nuts and bolts.
 - B) Bolting that is external of the cylinder uses standard materials; e.g. outer head-to-cylinder studs which are located outside the head gasket; pulsation bottle-to-cylinder bolting; etc.
 - C) ASTM A193, Grade B7M is used on certain cylinder designs where the head bolting is in the process gas passageway; e.g., Clamshell head designs.
- 3) Standard piston with rider bands or combination rings
- 4) Piston rods shall be annealed 4142 carbon steel (18-22 HRC) with Siemens Energy TC3 coating in the area of packing travel. (If the gas also contains CO₂, please refer to "GFP-16". If stainless steel rods are required, they will be made out of age hardened 17-4 PH material and TC3 coated.)
- 5) Suction port oilers are recommended. Alternatively, the valves can be supplied with special spring material.
- 6) Siemens Energy standard nodular iron valves will be furnished when pressures allow.
- 7) If forged steel cylinders are required for the application, a modified AISI 4132 steel alloy is required.

EXCEPTIONS TO NACE MR0175-94 SPECIFICATION:

- 1) A combination of Siemens Energy field experience and reference to specifications NACE MR0-175-94 and API 11P has been used to compile this GFP. Siemens Energy has taken into account stress levels, type of stress, exposure to sour gas and the critical nature of the part.
- 2) Hardness of Carbon and Low-Alloy Steels (3.2), and ferritic stainless steels (3.6) materials shall not exceed 22 HRC.
 - A) Tube Fittings. Tube fittings will be type 316 stainless steel double ferrule type. The hardness of the ferrule will exceed 22 HRC. Field experience with these fittings has been excellent.
 - B) Chrome Silicon Springs are not listed by NACE as an acceptable material. Siemens Energy has had favorable experience with this material in compressor valve service.
 - C) AISI 440C Stainless Steel hardened and tempered to 50 HRC. This material is used as a guide ring on PF valves. The stress level is minimal. Siemens Energy has had favorable experience.
 - D) Steel Nut, PF Valve. The stress level is low and Siemens Energy has favorable experience.
 - E) Rollpin. This is used as a valve plate guide on the PF valve. The stress level is low. Siemens Energy has favorable experience in lubricated valve service.
 - F) Piston rods made from AISI 4142 alloy steel are tempered back to 22 HRC or less. After the tempering heat treatment, threads are rolled on the rods, causing hardness at the roots of the threads to exceed 22 HRC. However, field experience in sour gas has been very good due to favorable residual stress pattern, surface finish, and grain flow patterns.

- 3) Siemens Energy has favorable experience with the following cast irons:
 - A) ASTM A278 Class 40 (Gray Iron), ASTM A536 (Ductile Iron) (Grade 65-45-12) and ASTM A395 (Ductile Iron) (Grade 60-40-18) for compressor cylinders and other components.
 - B) Continuous Cast Gray Iron is used on solid pistons and some small cylinder heads. It is equivalent to a Class 30/35 iron.
 - C) Centrifugally Cast Gray Iron is used for cylinder liners. It is equivalent to a Class 35 iron.
 - D) NiResist ASTM A436 Type 5 with 34-36% Ni (Minovar) is used as a piston rod collar on aluminum pistons due to its low coefficient of thermal expansion. The material is under compression at all times.
 - E) Ductile Iron, Continuous Cast Grade 80-55-06 is used for fully machined low to medium pressure valve seats and stopplates & guards.
 - F) Ductile Iron Castings Grade 100-70-30 is used for low to medium pressure PF valve seats and stopplates & guards.
- 4) COMMENTS TO NACE MR0175 SPECIFICATION:
 - A) Paragraph 6.2.2.1, ASTM A193, Grade B7M bolting 80,000 psi minimum yield strength and HRC 22 maximum hardness. This material is used on certain cylinder designs where the head bolting is in the process gas passageway; e.g., Clamshell head designs. It is also used as an option for cylinder bolting at an extra cost. This bolting meets the requirements of this Paragraph, however, it may exceed the 90,000 psi maximum yield strength limitation as outlined by API 11P, Paragraph 15.3.2.2. AISI 4142 alloy steel piston rods may also exceed the 90,000 psi yield strength limitation up to a maximum of 95,000 psi.
 - B) Paragraph 10.7.2, Aluminum Alloy 335, Temper T-7 (ASTM B-26). Siemens Energy uses either Alloy 355-T7, Alloy SR 319.0-T51, or Frontier 40H for aluminum pistons.

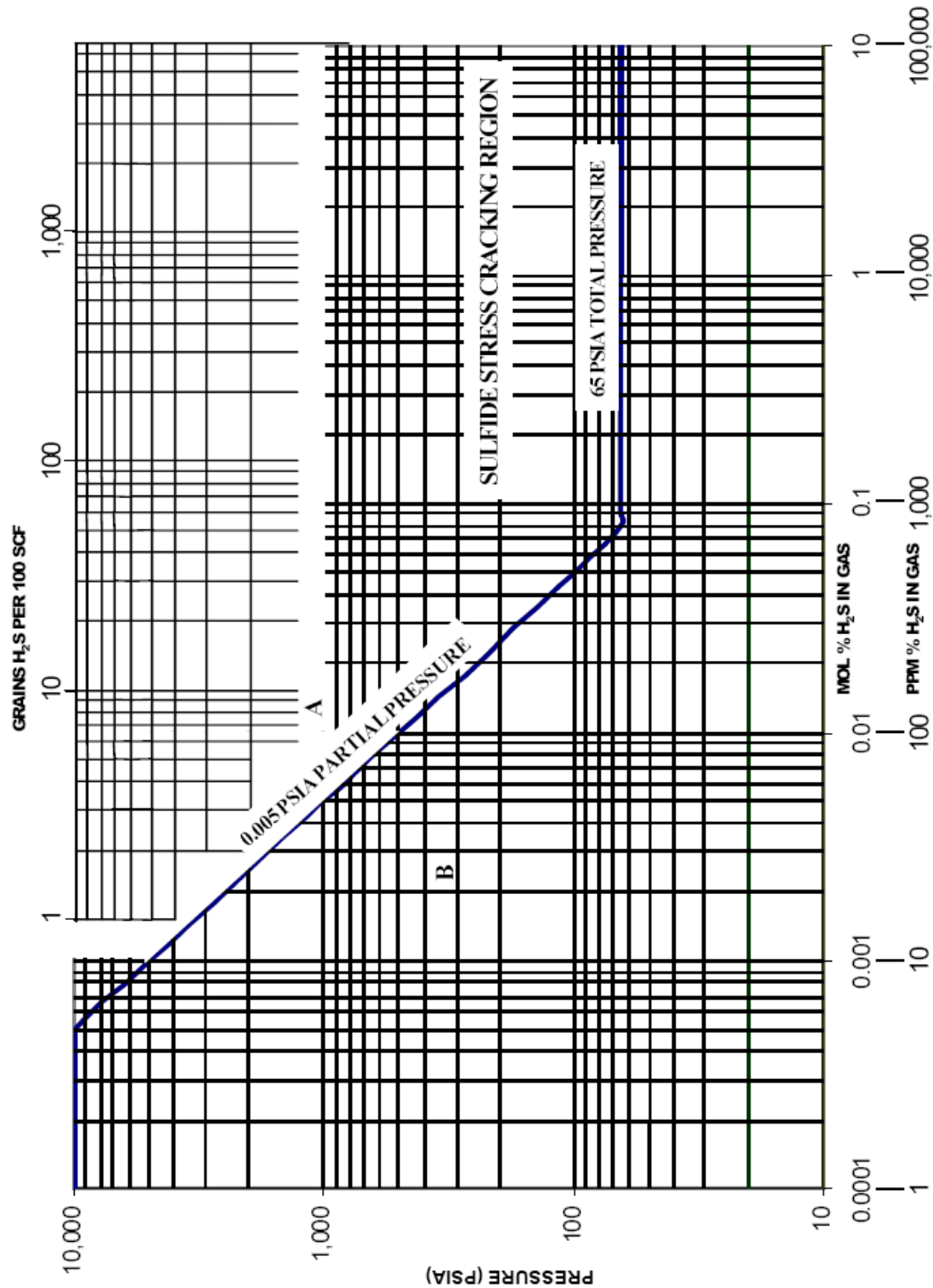
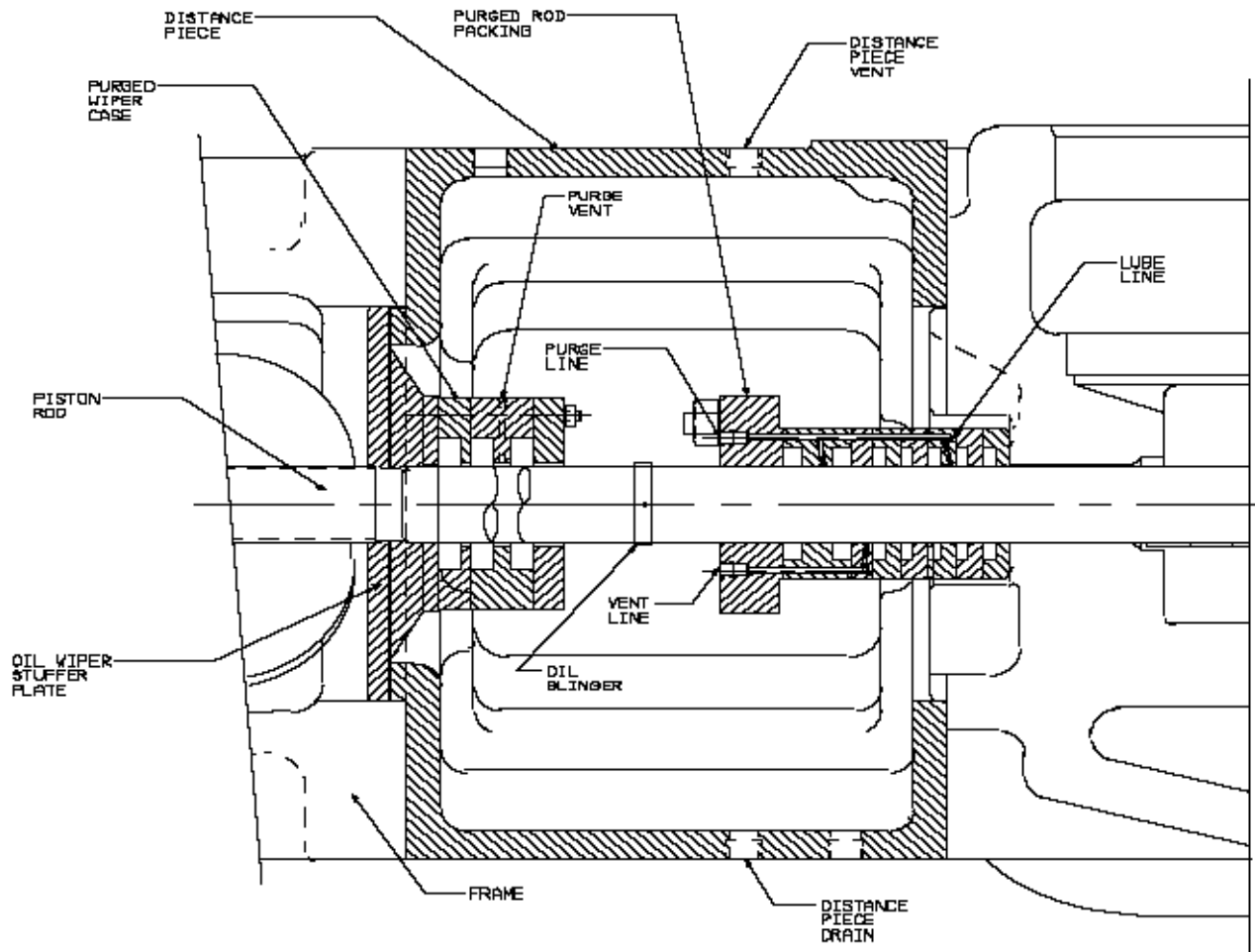


FIGURE 1

NOTE:
GAS A = SOUR GAS - THIS GFP APPLIES
GAS B = NOT SOUR GAS - THIS GFP DOES NOT APPLY


FIGURE 2

PURGED PACKING / PURGED WIPER ARRANGEMENT FOR SOUR GAS APPLICATIONS

1. **PURGED ROD PACKING** - In our "standard" rod packing, the vent line is just ahead of the last packing cup (closest to the frame). For sour gas service, the packing rings of the last two cups are exchanged for "side loaded" rings, with a PURGE line in between. This is so that any PURGE gas leakage will be in both directions (into the VENT line and into the distance piece), and any normal PROCESS gas leakage will be carried away by the VENT line.
2. **PURGED WIPER CASE** - For an extra measure of safety, two sets of "side loaded" rings, with a PURGE line in between, are added to the wiper case (at the frame end of the distance piece). Any leakage of PURGE gas will be in both directions (into the distance piece and into the frame extension).
3. **DISTANCE PIECE** - Any leakage into the distance piece, of either PURGE or PROCESS gas, will be carried away by the distance piece VENT line. An "oil slinger" (if required per the order) can be attached to the piston rod to keep any oil which has had contact with the PROCESS gas (from the rod packing) from traveling along the piston rod and into the frame. This may require an extra long distance piece.

This distance piece configuration creates a double buffer between the PROCESS gas and the frame. This arrangement produces the same effect as the API 11P Type 3 distance piece option.

Both PURGE gas lines should be protected from back pressure by a check valve outside the distance piece, supplied by the Reseller.